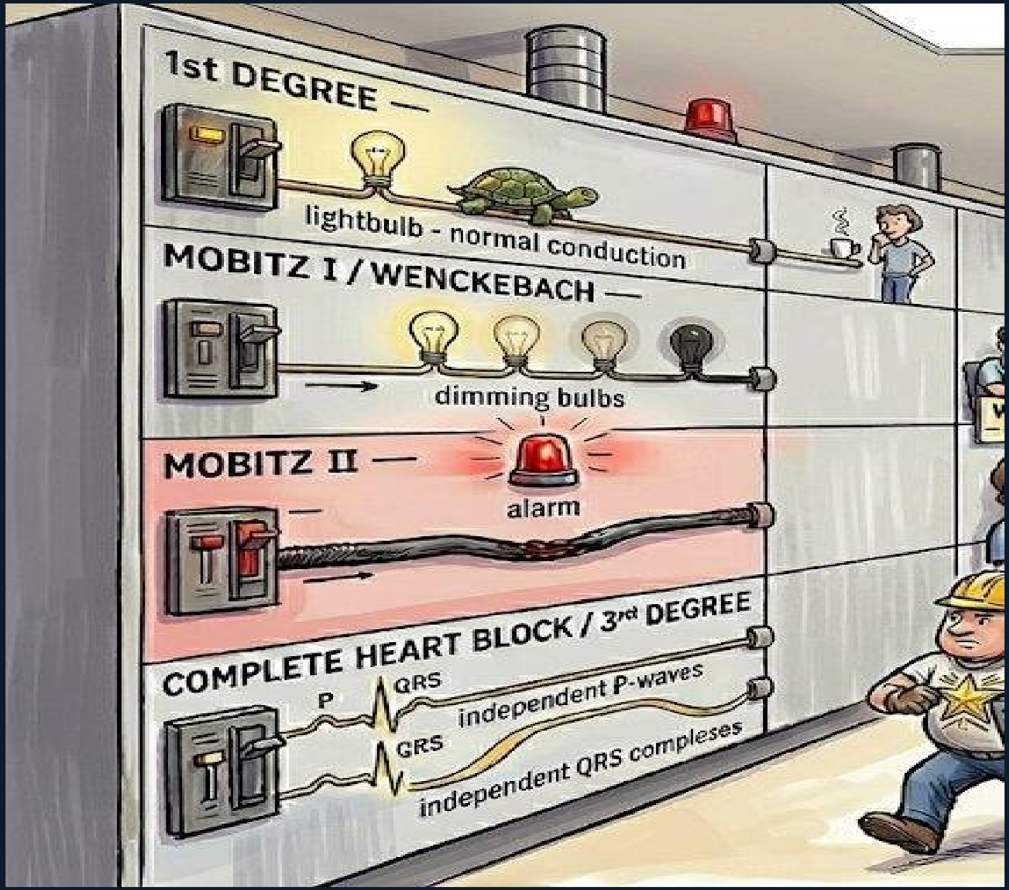
Scene B: AV Blocks & Pacemaker Indications

Electrical Substation · MUST MEMORIZE · 4 Board Questions

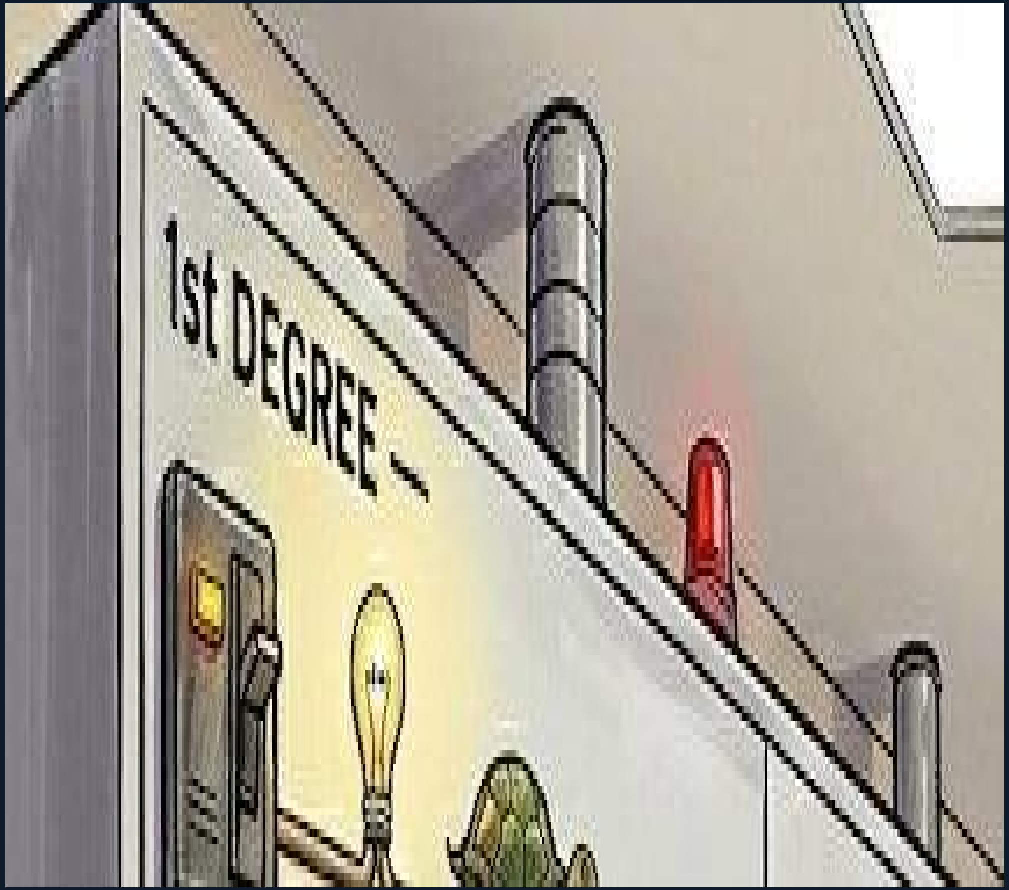


The AV Block Panel

The left wall shows four circuit breaker rows — 1st Degree, Mobitz I, Mobitz II, and Complete Heart Block — each depicting progressively more dangerous conduction failure.

This spatial layout lets you instantly classify AV blocks by ECG and know the correct action for each. The most important skill the boards test: distinguishing Mobitz I (observe) from Mobitz II (pacemaker always).

Board framework: before answering any AV block question, identify the row — does the PR lengthen before the drop (Mobitz I) or drop suddenly from a fixed PR (Mobitz II)?

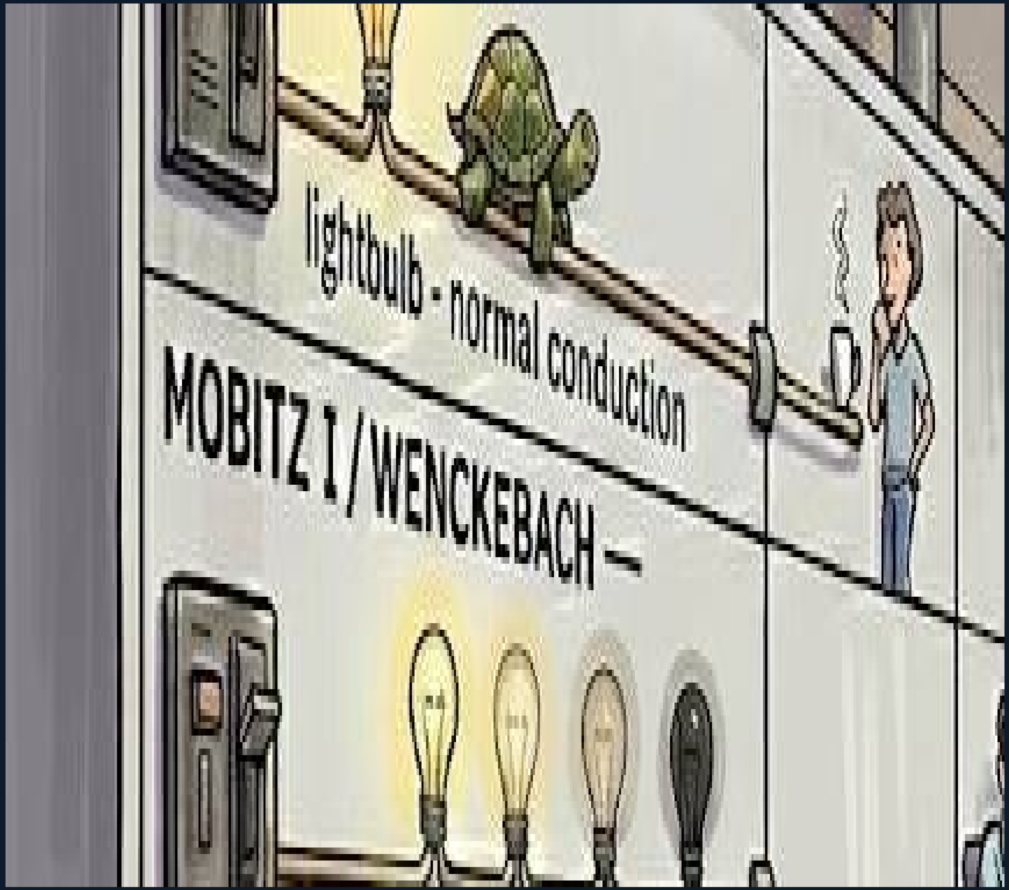


1st Degree AV Block

The top row shows an amber light and a tortoise walking along the wire — conduction is slow but every impulse gets through. A technician shrugs and sips coffee.

PR interval >200 ms but every P wave conducts to a QRS. No beats are dropped. This is delayed conduction, not blocked conduction. No treatment is needed.

Board trap: 'treat first-degree AV block with atropine' — wrong. Only treat if the patient is symptomatic from a low heart rate, and even then it is rarely due to first-degree block alone.

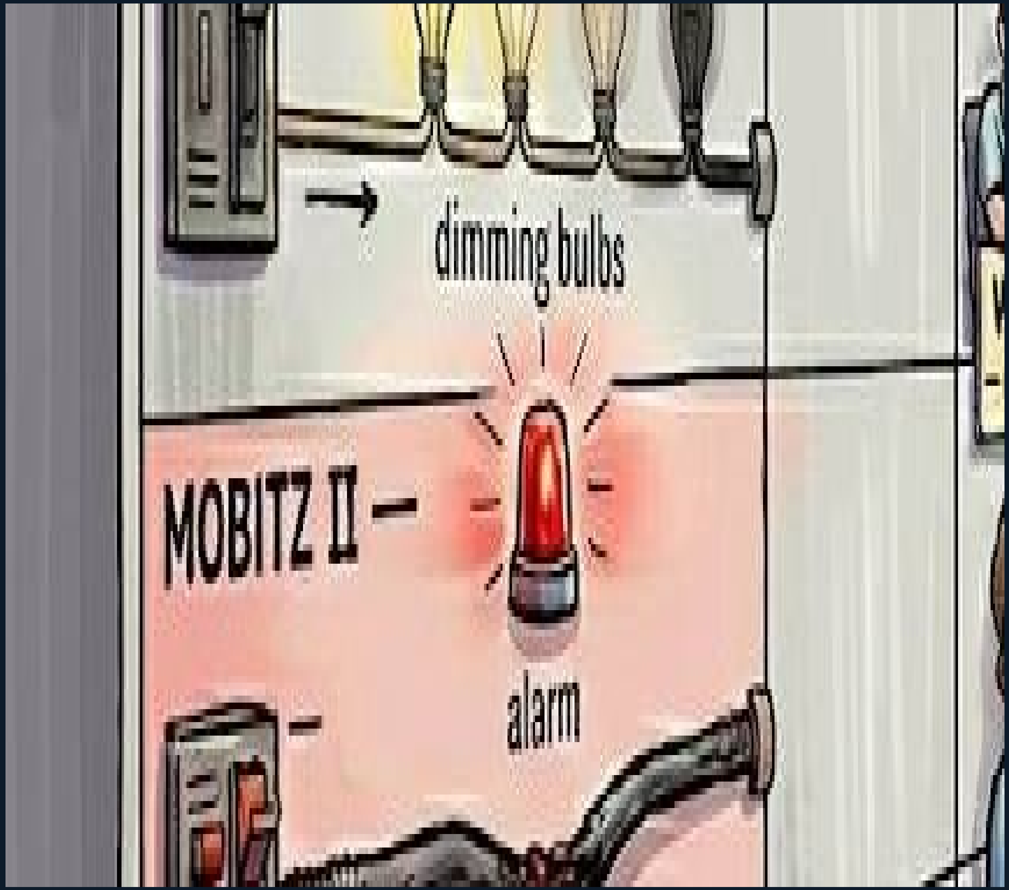


Mobitz I / Wenckebach

Three progressively dimming bulbs followed by one dark bulb — longer, longer, longer, drop. The WATCH/observe sign and relaxed technician encode the clinical response: observation.

Mobitz I: PR interval lengthens progressively until one QRS is dropped, then the cycle resets. Site: AV node. Cause: increased vagal tone, inferior MI, AV nodal blocking drugs. Generally benign.

Board trap: 'Mobitz I always needs a pacemaker' — wrong. Observe unless symptomatic. Exception: Mobitz I in the setting of progressive neuromuscular disease — pacemaker regardless.

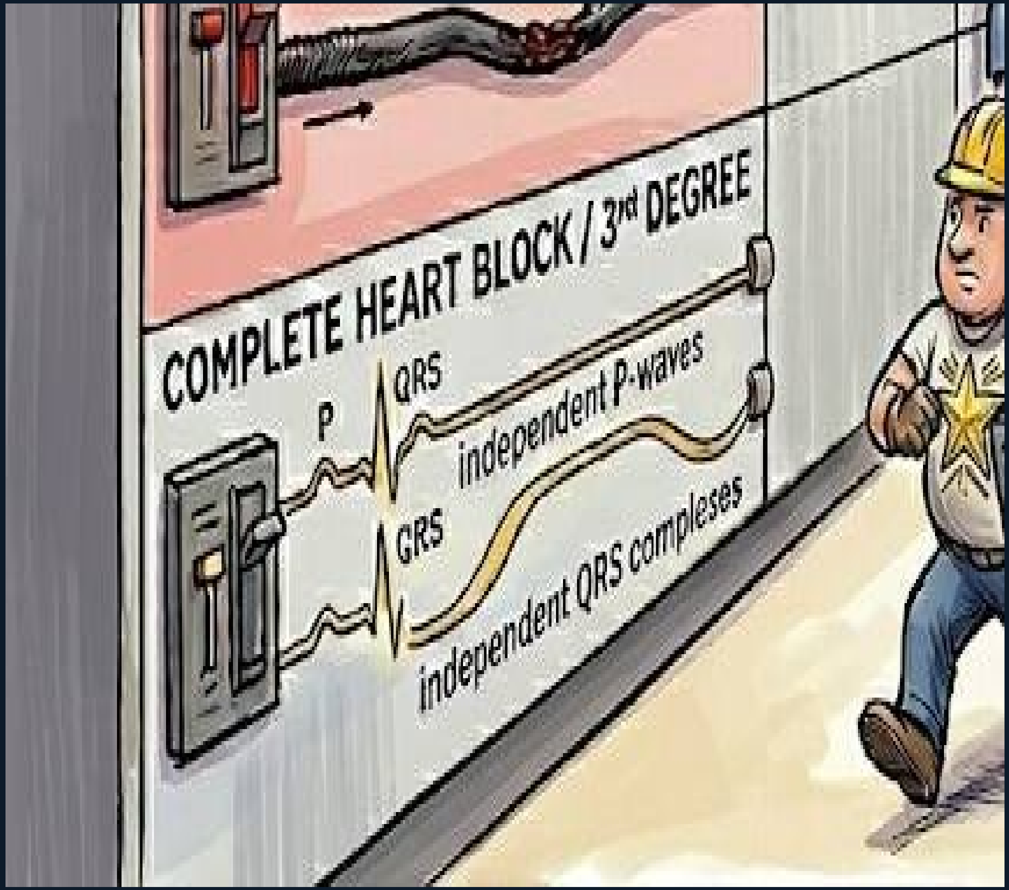


Mobitz II — Pacemaker Always

The Mobitz II row has a flashing red alarm, a sudden blackout, and a damaged infranodal wire. The gold-star pacemaker installer rushes to this panel. 'PACEMAKER — ALWAYS' is stamped on the breaker.

Mobitz II: constant PR interval, then a sudden dropped QRS with no warning. Site: infranodal (His-Purkinje). Often associated with wide QRS. Dangerous because it can progress unpredictably to complete heart block.

Board trap: 'observe Mobitz II' — wrong. Pacemaker is always indicated regardless of symptoms. Do not wait.

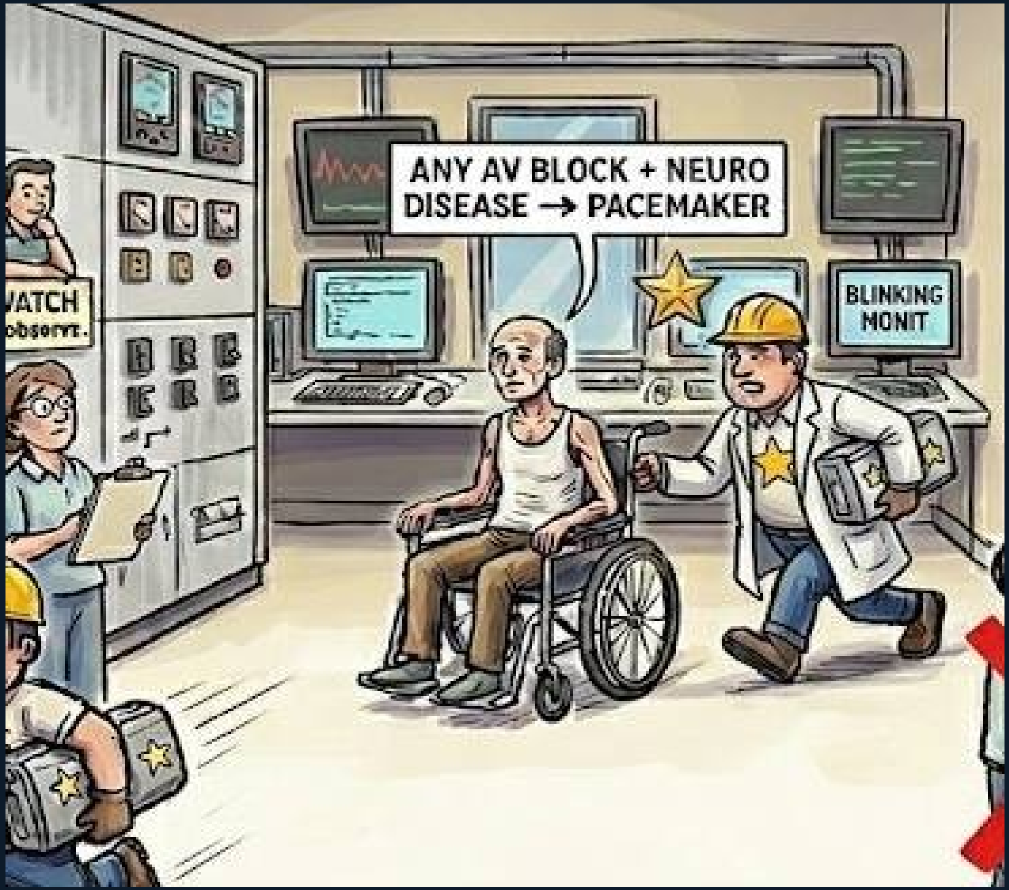


Complete Heart Block

Two parallel wires with no connection — P waves and QRS complexes march independently. The ventricular escape rhythm is labeled '20–40 bpm WIDE ESCAPE.'

Complete (3rd degree) AV block: no atrial impulses conduct to the ventricles. The ventricle generates its own escape rhythm — junctional (narrow, 40–60) or ventricular (wide, 20–40). Ventricular escape is more dangerous.

Board trap: 'atropine is definitive for complete heart block' — wrong. Atropine is a bridge. Permanent pacemaker is the definitive treatment.



Neuromuscular Disease Rule

A wheelchair patient with muscle wasting (neuromuscular disease) is getting a pacemaker despite having only Mobitz I on the panel. Sign: 'ANY AV BLOCK + NEURO DISEASE → PACEMAKER.'

In progressive neuromuscular diseases (myotonic dystrophy, Kearns-Sayre syndrome, Erb's muscular dystrophy), cardiac conduction defects progress inevitably. Any degree of AV block in these patients — even Mobitz I, even asymptomatic — warrants permanent pacemaker implantation (Class IIa).

Board trap: 'asymptomatic Mobitz I in myotonic dystrophy — observe' — wrong. Neuro disease overrides the usual 'observe' rule.

Acute Management Ladder

A four-rung ladder on the right wall: Rung 1 = ATROPINE, Rung 2 = TRANSCUTANEOUS PACING, Rung 3 = DOPAMINE/EPINEPHRINE, Rung 4 = TRANSVENOUS PACING. The permanent pacemaker waits at the top.

For acute unstable AV block (symptomatic bradycardia from block), this is the sequence. Atropine (0.5 mg IV) is first — works for AV nodal blocks (Mobitz I, 1st degree) but unreliable for infranodal (Mobitz II, CHB). Transcutaneous pacing buys time. Transvenous pacing is the bridge to permanent pacemaker.

Board trap: 'atropine reliably treats Mobitz II' — wrong. Atropine works at the AV node level; Mobitz II is infranodal.

ACE-I — Wrong Answer

A small figure holding an ACE-inhibitor bottle has a large red X stamped on it. A clipboard labeled 'OBSERVE' pointed at the Mobitz II row also has a red X.

ACE inhibitors have no role in treating AV block. They are tested because they commonly appear in HF management questions — students may default to them when managing a patient with both HF and AV block.

Board shortcut: AV block management = atropine (bridge) or pacemaker (definitive). No antihypertensives, no beta-blockers, no digoxin as treatment.

Scene B — Board Trap Master Summary

1st degree → no treatment

Delayed conduction; no dropped beats

Mobitz I → observe

Progressive PR then drop; AV nodal; benign

Mobitz II → pacemaker ALWAYS

Sudden drop, fixed PR; infranodal; dangerous

Complete heart block → pacemaker

AV dissociation; ventricular escape 20–40

Neuro disease + any AV block → PM

Class IIa even if asymptomatic Mobitz I

Atropine = bridge only

Works at AV node; unreliable for infranodal blocks

ACE-I = wrong answer

No role in AV block management

'Observe Mobitz II' = wrong

Always needs pacemaker regardless of symptoms

Infranodal = wide QRS clue

Wide QRS in Mobitz II → His-Purkinje disease

Transcutaneous → transvenous → PM

Escalation sequence for acute unstable block